Yuetang Liang

Preferred Name: Michelle

University of Ottawa

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Objective

I am in Honoured Computer Science first-year student at the University of Ottawa with a strong foundation in mathematics, programming, and logical problem-solving. I am seeking a co-op position in IT/Network Security where I can apply my analytical skills and technical knowledge to contribute to secure network operations, vulnerability assessments, and system security research. I am eager to gain hands-on experience in a Security Operations Center environment and to apply my skills to practical cybersecurity challenges.

Education

University of Ottawa – Ottawa, ON
Honoed Computer Science (Co-op) Undergraduate Program
Expected Graduation: June, 2029

Relevant Courses: Network Security & Cryptography, Systems Programming & Algorithms, Linear Algebra, Probability and Statistics for Computing, Data Structures and Software Design, Risk Assessment & Ethical Hacking

Academic Achievement:

Completed over 1000 **LeetCode problems**, covering topics such as data structures, algorithms, and logic-based problem solving.

Achieved proficiency in solving **medium to hard-level coding challenges**, demonstrating analytical thinking and systematic problem-solving skills.

Technical Skills

- **Programming and Scripting:** Python (automation, network tools, data analysis), Bash, PowerShell, Java, C++
- **Mathematical and Analytical Tools:** Linear algebra, discrete mathematics, logic, combinatorics, probability modeling
- **Networking and Security Tools:** Wireshark, Nmap, OpenVAS, Nessus, Snort, Wireless Intruder Detection Systems (WIDS)
- **Operating Systems and Virtualization:** Linux, Windows Server, macOS, VMware, VirtualBox
- **Other Skills:** Git, Firewalls, VPNs, System Hardening, Security Auditing

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Experience

Cybersecurity & Network Security Research – University of Ottawa, Ottawa, ON
[September 2025 – December 2025]
As part of my research work within university communities and computer security labs, I investigated IT and network security practices using controlled, authorized environments. I worked with campus research networks, lab-managed systems, and small simulated LAN setups to study how network configurations affect security. I used tools such as **Wireshark** to analyze packet flows and detect abnormal traffic patterns, **Nmap** to map lab devices and verify exposed services, and **OpenVAS/Nessus** to identify vulnerabilities in test systems. I also developed **Python and Bash automation scripts** to streamline scanning tasks, organize logs, and perform repeatable assessments.

My goal in using these tools was to enhance the reliability of security evaluations, reduce manual workloads during lab research, and ensure consistent testing across different university network scenarios. These activities helped identify configuration weaknesses, insecure services, and data exposure risks within controlled environments, ultimately strengthening security awareness and contributing to safer operational practices within university labs and research communities.

IT Support and System Assistance – University of Ottawa, Ottawa, ON
[September 2025 – December 2025]
I assisted in configuring and maintaining secure network systems for departmental operations. I applied programming techniques to automate routine IT tasks, streamline workflows, and reduce manual effort. I monitored network activity and used analytical reasoning to troubleshoot technical issues efficiently. This role strengthened my adaptability, teamwork, and ability to respond to unexpected challenges in a fast-paced IT environment.

Academic Project: Secure Network Simulation – University of Ottawa
[September 2025]
Designed and evaluated a secure network environment for a university club within controlled lab and campus settings. Used tools such as Wireshark, Nmap, and OpenVAS in authorized test environments to analyze traffic behavior, assess vulnerabilities, and verify proper service

configurations. Developed Python/Bash automation scripts to perform repeatable scans, organize logs, and improve assessment accuracy. Provided recommendations to enhance security posture, protect member data, and ensure reliable network operations for club activities.

Certifications

- Ready for CompTIA Security+ (in progress)
- Ready for Cisco CCNA (in progress)
- Ready for Certified Ethical Hacker (CEH) (in progress)

Core Competencies

I have a strong foundation in logical reasoning and problem-solving, combined with programming and mathematical skills. I am adaptable, eager to learn, and capable of approaching complex technical challenges methodically. I can work effectively in teams, communicate technical concepts clearly, and apply my knowledge to analyze and improve system security.

Relevant Coursework & Academic Achievement

ITI1120 – Introduction to Computing I
I have achieved strong results by mastering fundamental programming concepts, including algorithm design, data structures, recursion, and problem decomposition. Demonstrated the ability to write clean, well-structured Python code and apply logical thinking to solve increasingly complex computational problems.

MAT1341 – Introduction to Linear Algebra
I am excelled in vector spaces, matrices, Gaussian elimination, orthogonality, projections, and least-squares approximation. Applied rigorous mathematical reasoning to solve multi-step algebraic problems and connect linear algebra concepts to computer science applications such as graphics, optimization, and data analysis.

MAT1320 – Calculus I
I have earned high academic standing through strong conceptual understanding of limits, derivatives, optimization, and mathematical modeling. Applied systematic problem-solving techniques and demonstrated precision in symbolic manipulation, logical reasoning, and multi-stage calculations.

The following results for reference:

MAT1341 E00 Intro. To Linear Algebra [LEC] 20259

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Grades

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Grade Item	Points	Weight Achieved	Grade	Comments and Assessments
Midterm 1	37 / 40	13.88 / 15	A+	
Midterm 2	38 / 40	14.25 / 15	A+	
Midterm 3	34 / 40	12.75 / 15	A	

MAT1320 B00 Calculus I [LEC] 20259

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Grade Item	Points	Weight Achieved	Grade	Comments and Assessments
Getting to know Möbius	- / 5	- / 0	-	
Assignment 1	10 / 10	0 / 0	A+	
Assignment 2	10 / 10	0 / 0	A+	
Assignment 3	10 / 10	0 / 0	A+	
Assignments 1, 2, 3 - Practice Version	- / 30	- / 0	-	
Assignment 4	10 / 10	0 / 0	A+	

Grades

Print

Grade Item	Points	Weight Achieved	Grade	Comments and Assessments
Quizzes		5 / 5	A+	
Quiz 1	5 / 5	2.5 / 2.5	A+	
Quiz 2	22 / 22	2.5 / 2.5	A+	View Quiz Attempts
Assignments		18 / 18	A+	
Assignment 1	80 / 80	9 / 9	A+	
Assginment 2	100 / 100	9 / 9	A+	
Assginment 3	- / 100	- / -	-	
Assginment 4	- / 100	- / -	-	
Midterm	23 / 25	0 / 0	A+	
Midterm adjusted	25 / 25	25 / 25	A+	